

Assignment 1

Statistical Learning Theory and Data Science

Course Code: BI3424/DS3214

Submitted by *Achyusman Samanta (Roll no: 20211147)*

Results and Observations:

These results and plots are obtained with 100 different datasets each having $N = 3$ points.
The target function used to sample the data points:

$$y = (x - 2)^2 + 4 + \text{noise} \sim \text{Normal}(0, 0.2)$$

Hypothesis That best approximates f in the average sense:

The constant hypothesis that best approximates f in the average sense: $y = 42.7869$
The linear hypothesis that best approximates f in the average sense: $y = -3.8764x + 25.7266$

Graphs:

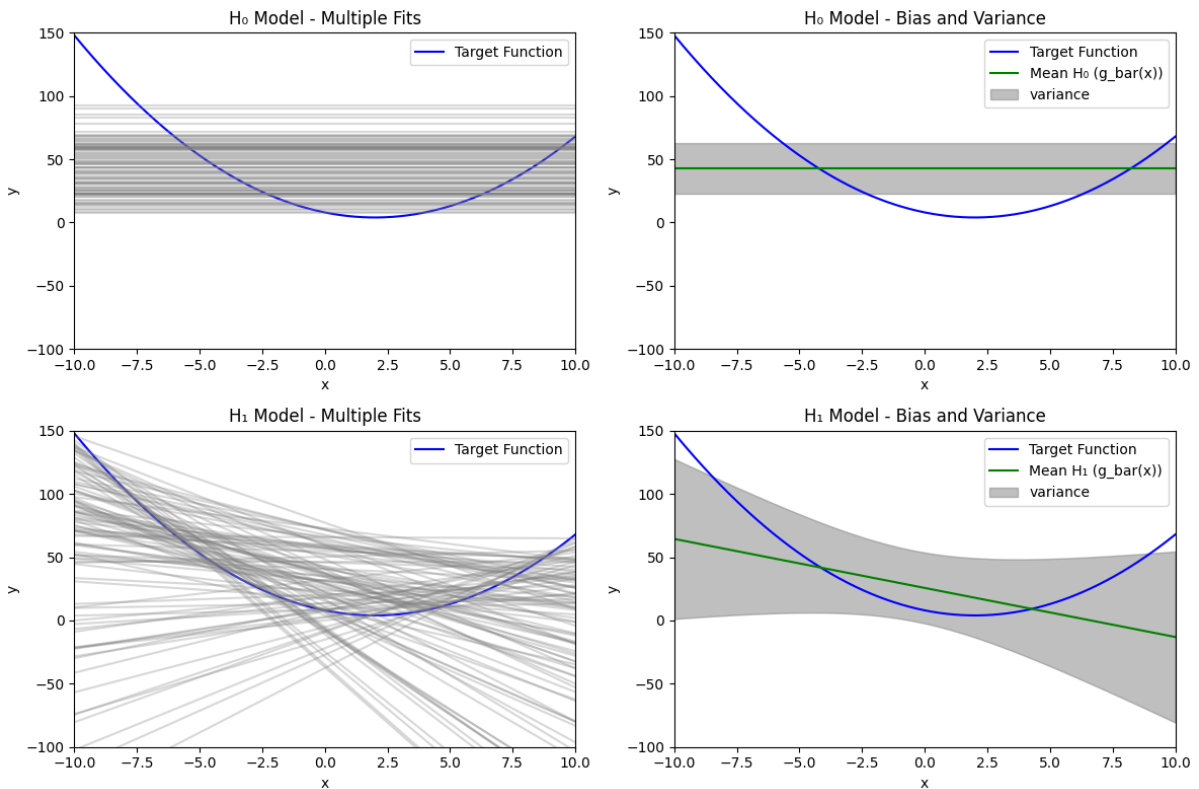


Figure1: The two plots on the left side of the figure show the resulting fits on the same randomly sampled data sets for H_0 (left-upper) and H_1 (left-lower) models. The two plots on the right side of the figure show the average hypothesis ($\bar{g}(x)$) with variance indicated by the grey shaded region in the form of $\bar{g}(x) \pm \sqrt{\text{var}(x)}$ for H_0 (right-upper) and H_1 (right-lower) models.

Bias and Variance components:

```
Bias and Variance components:  
H0 (Constant Model) - Bias: 1428.7733, Variance: 400.2454, E_out: 1829.0187  
H1 (Linear Model) - Bias: 1138.6173, Variance: 1953.2849, E_out: 3091.9022
```

Conclusion:

We determine the appropriate hypothesis class based on their expected out-of-sample errors (E_{out}). E_{out} is obtained by adding the bias and the variance terms. Bias measures how far the average model's predictions are from the true target function while variance is the measure of the variability of the predictions across different data sets.

With the linear model (H_1), the learned hypothesis is wilder and varies extensively depending on the data set. Moreover, although the average hypothesis seems to be a reasonable fit with a relatively smaller bias, large variability leads to a much higher variance resulting in a larger expected out-of-sample error. On the other hand, with the constant model (H_0), which is a fairly simpler model the fits are much less volatile and we have a significantly lower variance and hence a smaller expected out-of-sample error even if the bias is a bit high. Therefore, **constant (H_0) is the preferred model given this situation** but this might change as the number of data points increases.

As per the question we used $N = 3$ to generate the above results but if we take a bigger data set, the bias term will be the dominant part of E_{out} , and in this case the linear model (H_1) will win. For instance, the results obtained with $N = 10$ is show below:

```
The constant hypothesis that best approximates f in the average sense:  $y = 40.8481$   
The linear hypothesis that best approximates f in the average sense:  $y = -4.2085x + 36.0861$   
  
Bias and Variance components:  
H0 (Constant Model) - Bias: 1427.1546, Variance: 157.5261, E_out: 1584.6807  
H1 (Linear Model) - Bias: 922.1393, Variance: 438.5682, E_out: 1360.7075  
  
Conclusion: Constant class of hypotheses( $H_1$ ) is more appropriate for prediction
```